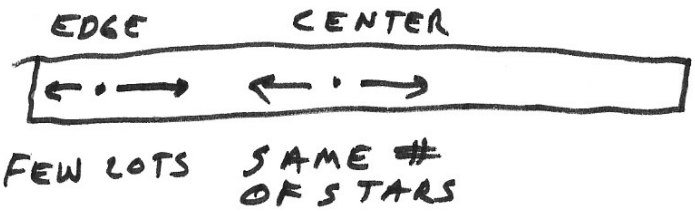
WHAT IS THE MILKY WAY?

ANCIENTS: EXACT NATURE IS UNKNOWN.
APPEARANCE:
- FAIRLY NARROW BAND
- WINTER AND SUMMER
- COMPLETELY ENCIRCLES THE SKY
- STRUCTURE (SUMMER)
- BULGES NEAR SAGITTARIUS

GALILEO: MADE OF "COUNTLESS MILLIONS" OF DIM STARS
(TODAY: 200 BILLION STARS)
MILKY WAY GALAXY (OUR GALAXY)
FLATTENED, DISK-SHAPED SYSTEM OF STARS
"MILKY WAY" STARS - VERY FAR AWAY
INDIVIDUAL STARS IN OUR SKY - MUCH CLOSER
BULGE → NUCLEAR BULGE

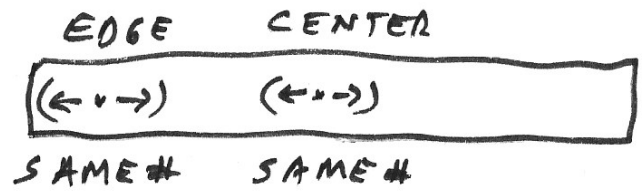
WHERE ARE WE / SOLAR SYSTEM / SUN?

1780s: WILLIAM HERSCHEL: STAR COUNTS



RESULTS: ROUGHLY SAME # OF STARS SEEN → CENTER!
NOPE!! STAR COUNTS WON'T WORK (DUST)

REALITY:



1920: HARLOW SHAPLEY

3D MAP OF THE DISTRIBUTION OF GLOBULAR CLUSTERS.

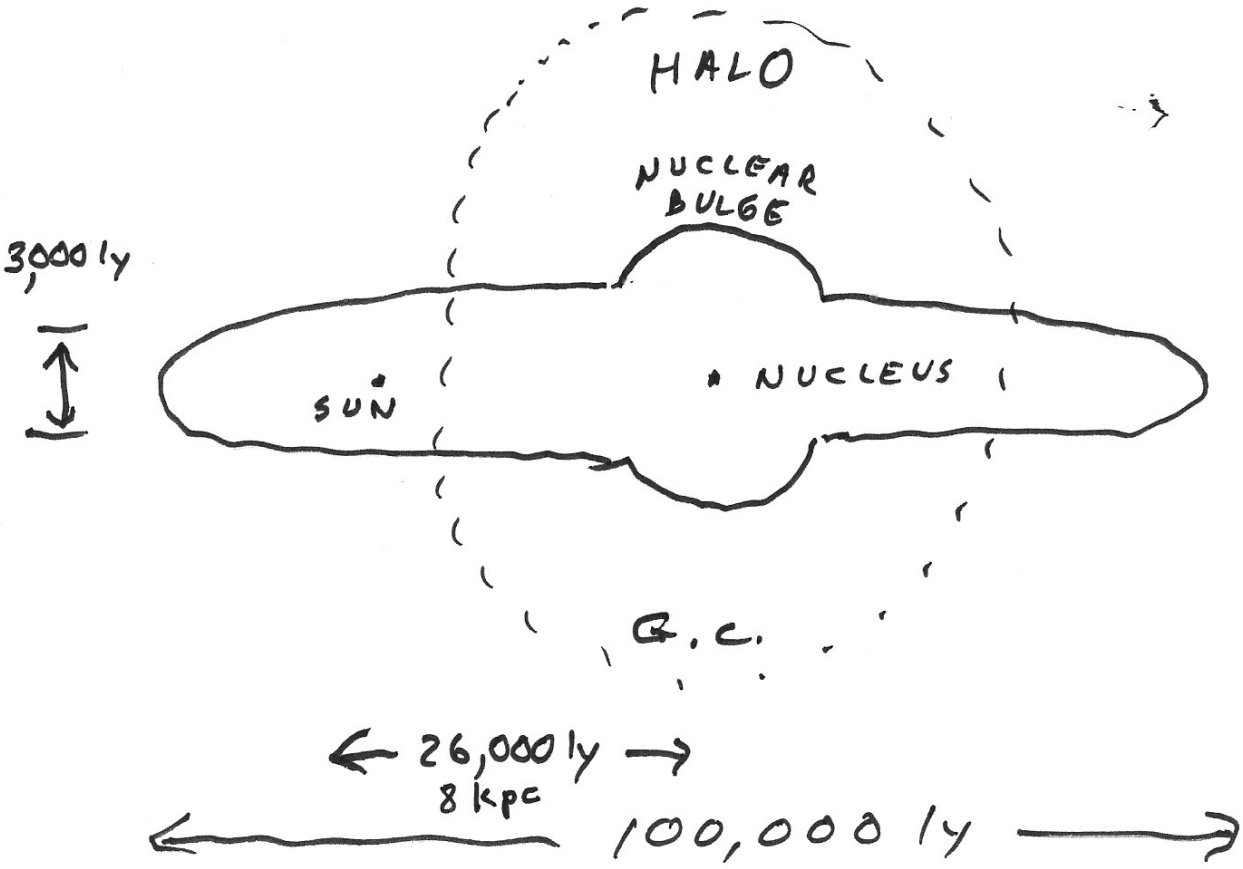
GET THEIR DISTANCES (CEPHEIDS)

ASSUMED THIS IS THE CENTER (NUCLEUS) OF THE GALAXY

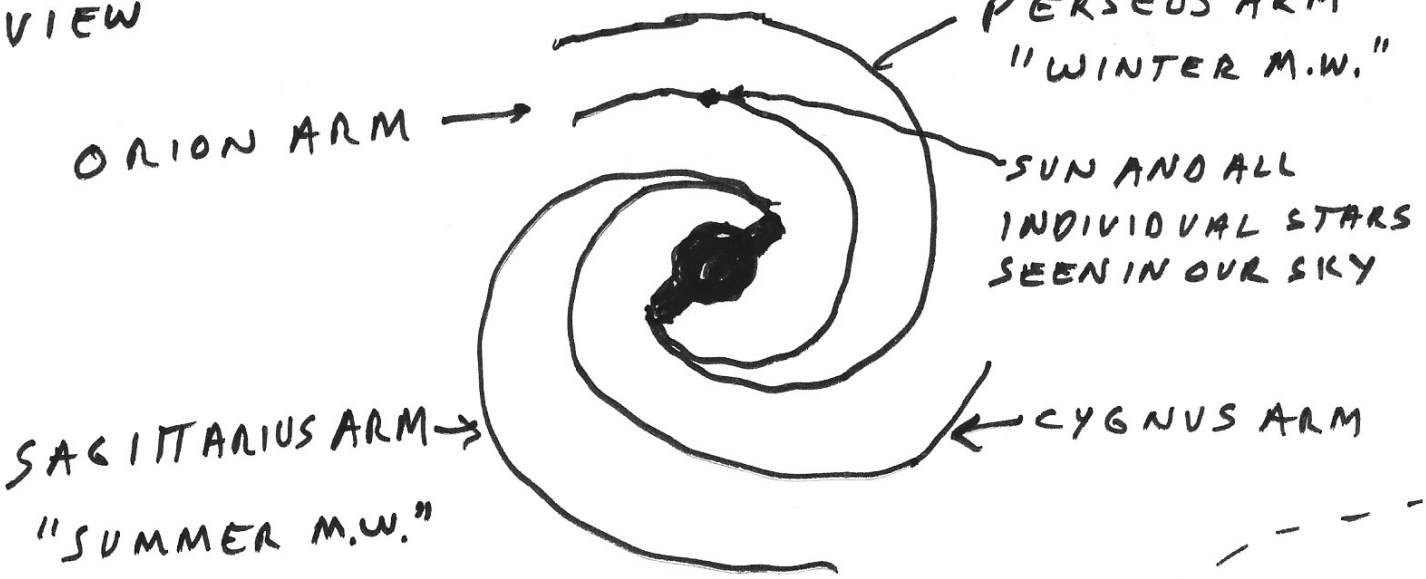
YEP!

GET THE STRUCTURE OF THE GALAXY? IR, RADIO* (21 CM SPIN-FLIP RADIATION)

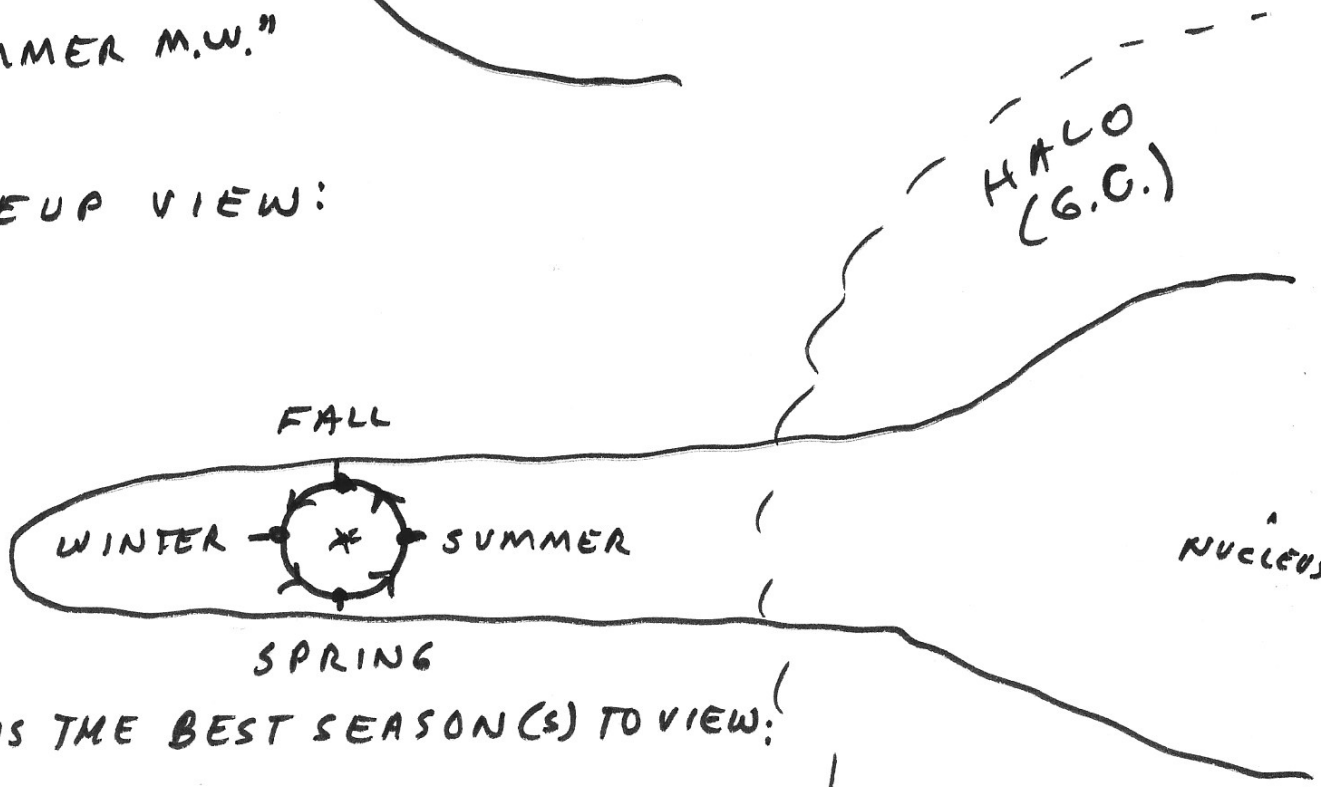
EDGE ON VIEW:



FACE ON VIEW



CLOSEUP VIEW:



WHAT IS THE BEST SEASON(S) TO VIEW:

1. G.C. : SUMMER
2. NEBULAE & O.C. : SUMMER & WINTER
3. OTHER GALAXIES : FALL & SPRING
   * VIRGO CLUSTER

ABOUT 90% THE GALAXY'S STARS RESIDE IN THE N.B. AND HALO.

FIRST APPROXIMATION:

ALL STARS IN THE GALAXY ORBIT THE NUCLEUS.

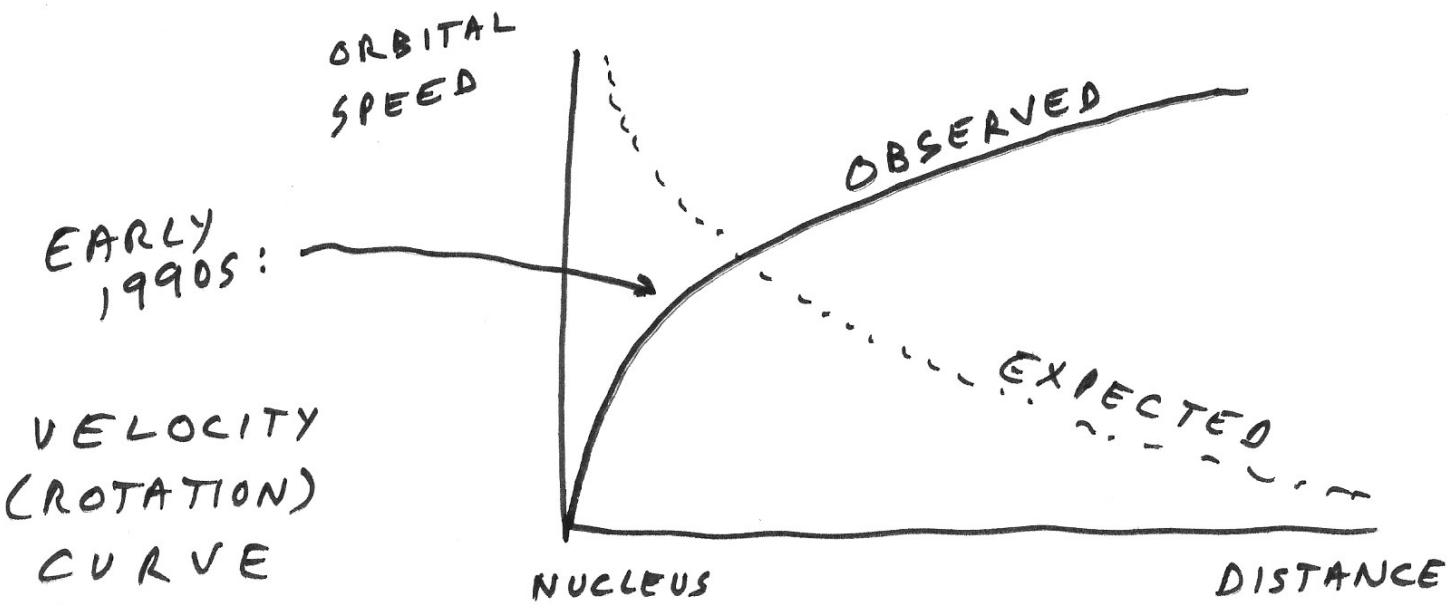
USING THE G.C. IN HALO AS A REFERENCE WE CAN DETERMINE THE SUN'S ORBITAL PROPERTIES:

SPEED: OVER ½ MILLION M.P.H.

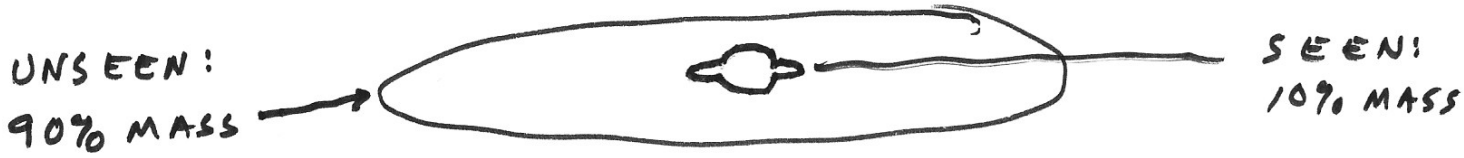
PERIOD: 230 MILLION YEARS (GALACTIC YEAR)

→ EARTH IS ABOUT 20 G.Y. OLD

NOW, GET THE ORBITAL PROPERTIES OF OTHER DISK STARS:



1990s: MOST (90%) OF THE GALAXY'S MASS IS IN A VERY VAST, EXTENDED HALO:



MISSING MASS → DARK MATTER!

2 POSSIBILITIES:

① WIMPS (Weakly Interacting Massive Particles)

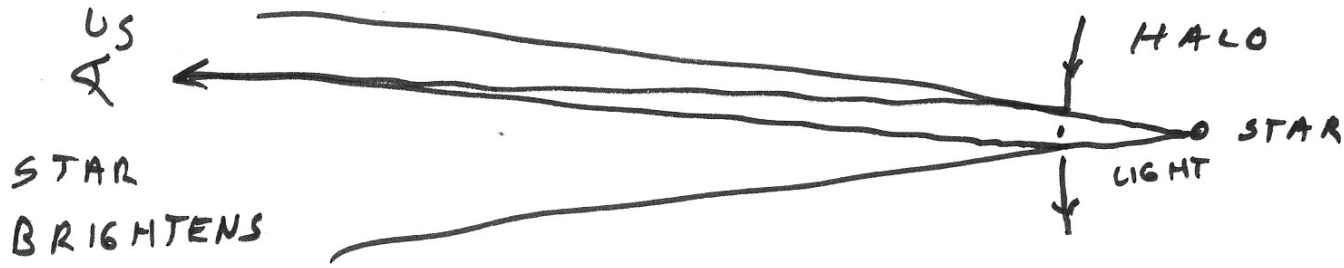
→ DIFFERENT FORM OF MATTER MOST LIKELY

VS.

② MACHOs (Massive Compact Halo Objects)

→ ORDINARY, NON-LUMINOUS MATTER

MICROLENSING EVENT:



WE SAW 1! BUT, STILL, 1 ☹

COSMOLOGICAL PRINCIPLE: NO SPECIAL PLACE EXISTS IN THE UNIVERSE.

ALL OTHER GALAXIES? 90% D.M. ✓

WHY SPIRAL ARMS? TIME →

SHOULD NOT EXIST! WINDING DILEMMA:

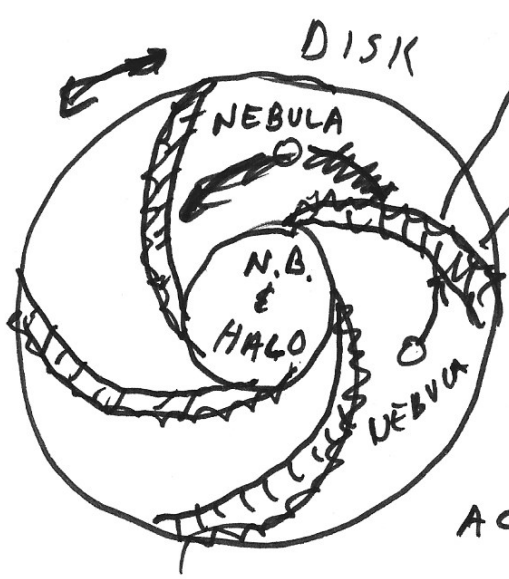
SOMETHING MUST MAINTAIN THIS SPIRAL STRUCTURE!

POSSIBLE ANSWERS < SPIRAL DENSITY WAVES
SELF-PROPAGATING STAR FORMATION

TODAY: MOST LIKELY BOTH.

FACE ON VIEW:

SHORT EXPOSURE:

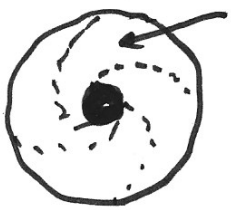


COMPRESSION WAVES

MOST STAR FORMATION IN THE DISK
→ MOST O STARS

SPIRAL DENSITY WAVES (O STARS) ACT LIKE "MALIBU LIGHTS"

LONG EXPOSURE:



TRAVEL ONLY A SHORT DISTANCE THEN THEY DIE
→ TYPE II SUPERNOVA

TRIGGERS MORE STAR FORMATION IN THIS REGION:

SELF-PROPAGATING STAR FORMATION

MOST LIKELY THIS IS THE ANSWER.

WHAT LURKS AT THE NUCLEUS OF THE GALAXY?

TRAVEL? NOPE! RELY ON OBSERVATIONS

VISIBLE LIGHT → CAN'T SEE IT

RADIO →

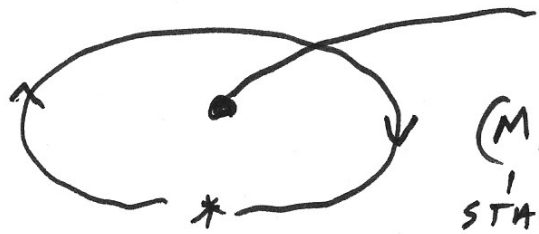
← SAGITTARIUS A

SAGITTARIUS A*

(NUCLEUS)

→ RADIO BRIGHT (STARS? NOPE!)

IR →



NUCLEUS: SEE NOTHING

$$(M_1 + M_2) P^2 = a^3$$

M_1: STAR, M_2: NUCLEUS

STAR (P, a)

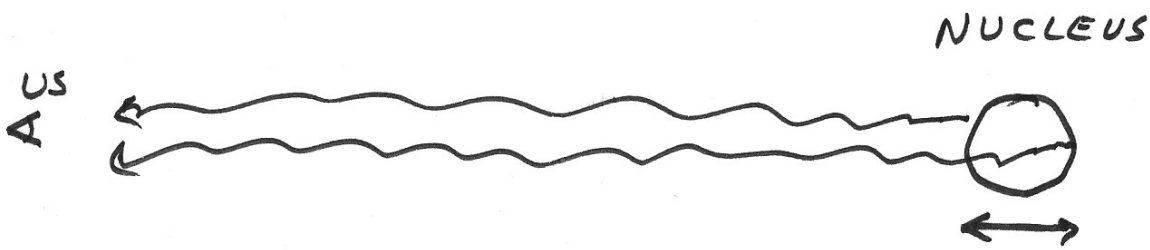
MASS OF NUCLEUS IS ABOUT 4 MILLION $M_\odot$!

MASSIVE GLOBULAR CLUSTER? NOPE!

UV → CAN'T SEE IT

X RAYS → BRIGHT (ENERGETIC)

γ RAYS → BRIGHTNESS VARIES IN AS LITTLE AS 1 DAY.



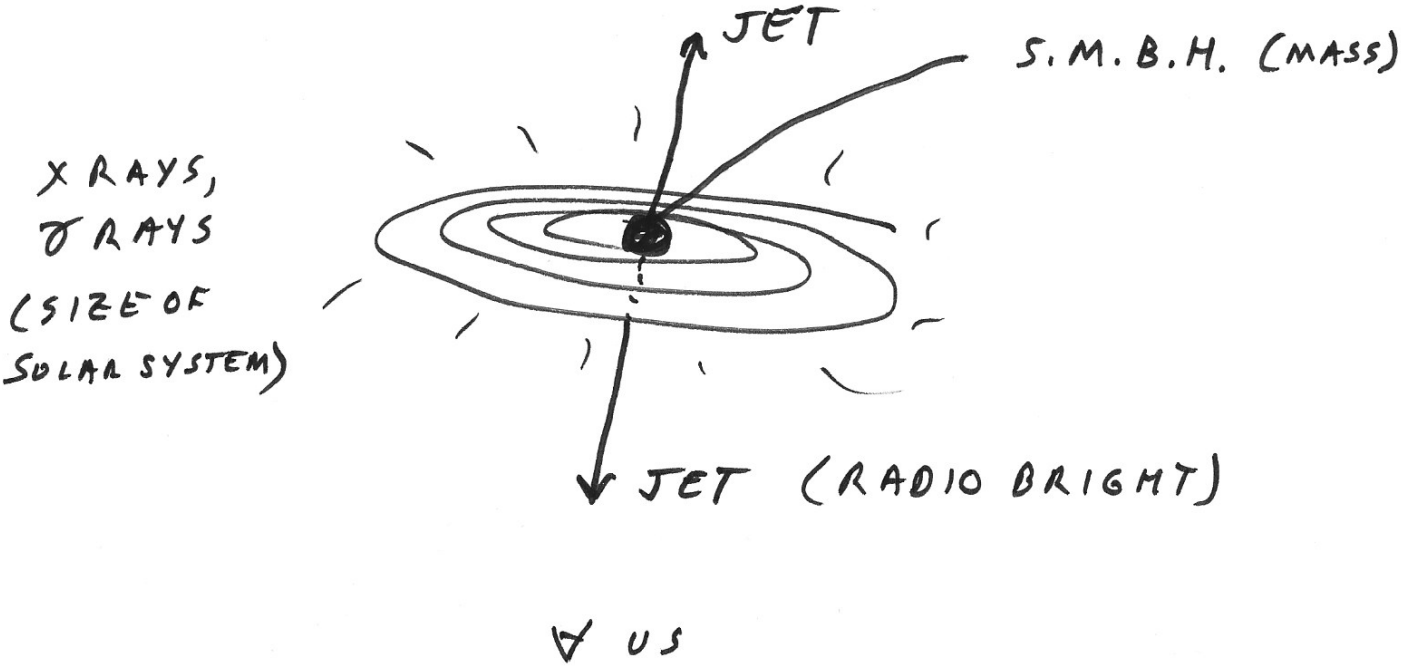
SIZE AS OUR SOLAR SYSTEM!

LIGHT TRAVEL TIME SIZE

→ 1 LIGHT DAY ACROSS

→ SUPERMASSIVE BLACK HOLE!

"CENTRAL ENGINE" MODEL:



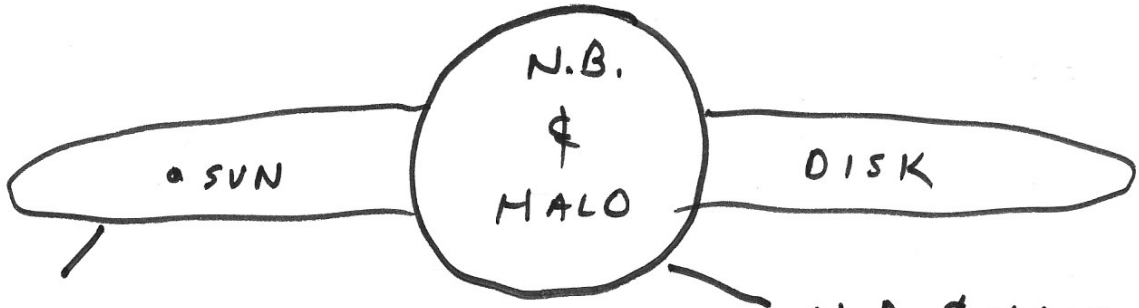
THIS IS MOST LIKELY WHAT IS AT THE NUCLEUS.

COSMOLOGICAL PRINCIPLE:

ALL OTHER MAJOR GALAXIES SHOULD ALSO HAVE A CENTRAL ENGINE.

H.S.T.: YEP!

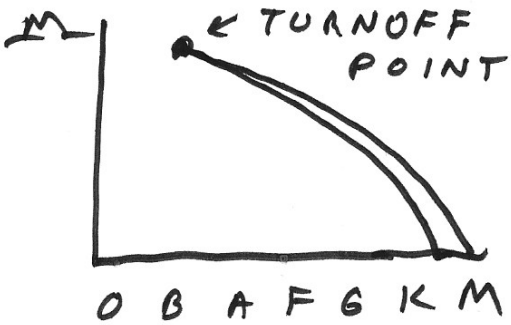
ALSO: AS MASS OF GALAXY ↑
S.M.B.H. MASS ↑



DISK

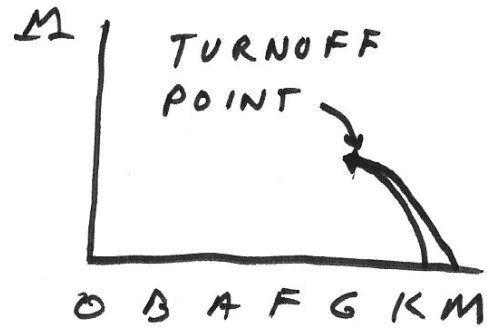
N.B. & HALO

OPEN CLUSTERS



100 MILLION YEARS OLD
→ YOUNG

GLOBULAR CLUSTERS



13 BILLION YEARS OLD
→ OLD

AGE OF A CLUSTER = M.S. LIFETIME OF A STAR AT THE TURNOFF POINT.

UNIVERSE IS MOSTLY H, He. ASTRONOMERS CALL ALL OTHER ELEMENTS "METALS."

SPECTRA OF STARS:

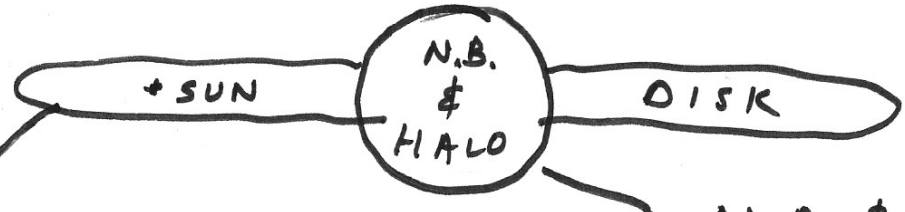
N.B. & HALO: "METAL POOR" STARS → POPULATION II STARS

DISK: LOTS OF "METAL RICH" STARS → POPULATION I STARS

SUN? POP I STAR

POP. II : FIRST GENERATION STARS

POP. I : RECYCLED STARS



DISK

CAN HAVE THESE

BUT ONLY HERE IS THERE NEBULAE, STAR FORMATION, YOUNG, HIGH MASS, O, B, POP I STARS.

→ BLUISH (ONGOING STAR FORMATION)

N.B. & HALO

(LITTLE VARIATION).

NO NEBULAE, NO STAR FORMATION, ONLY OLD, LOW MASS, G, K, M, POP. II STARS.

→ ORANGISH

GALAXIES

LATE 1700s: CHARLES MESSIER (COMET HUNTER)

WHEN FIRST DISCOVERED THEY LOOK LIKE "FAINT FUZZIES" IN TELESCOPES.

BUT, THERE ARE MANY FAINT FUZZIES THAT AREN'T COMETS (DEEP SKY OBJECTS)

COMPILES A LIST OF THE KNOWN ONES: M OBJECTS (110).

M 31 → ANDROMEDA NEBULA

MID 1800s: LORD ROSSE

"SPIRAL NEBULA"